gpeyt1-32 · Rock slope with joint water

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A sandstone cut contains a planar bedding joint dipping α = 38° (length L = 55 ft) and a vertical tension crack behind the crest. After a storm the tension crack fills to hw = 18 ft with water that drains along the joint to the face. Block weight W = 210,000 lb/ft. Joint c = 300 psf, φ = 32°. No seismic load. Rock RMR = 58 (fair).

Most nearly, the factor of safety with the crack full of water is:

A. 0.93
B. 0.85
C. 0.78
D. 0.70

Given equations / data

> FS = [cL + (W cos α − PWR − PWJ sin α) tan φ]/(W sin α + PWJ cos α); PWJ = ½γw hw²; PWR = ½γw hw L

Worked solution

1. PWJ = ½(62.4)(18)² = 10,109 lb/ft
2. PWR = ½(62.4)(18)(55) = 30,888 lb/ft
3. N′ = 165,482 − 30,888 − 6,224 = 128,371
4. Resisting = 16,500 + N′ tan32 = 96,715
5. Driving = 129,289 + 7,966 = 137,255
6. FS = 0.70 (dry 0.93)

Final answer

0.70

Why the other values are wrong

- **0.93** — ignored water forces
- **0.85** — omitted uplift PWR on the joint
- **0.78** — omitted the tension-crack force PWJ

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure, p. 104

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=8cX88YWUo4U, https://www.youtube.com/watch?v=jvEQeZuqydI (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.